

# SAO DEMO CONTROLLER V2

## FROM BADGE

SAO CLIENT PLUG  
<https://www.digikey.com/en/products/detail/adam-tech/BHR-06-VUA/10414837>

GPIO1 is also data from the badge to the SAO (Badge TX, SAO RX).  
GPIO2 is also data from the SAO to the badge (Badge RX, SAO TX).

## MICROCONTROLLER

RP2040-Zero  
<https://www.waveshare.com/wiki/RP2040-Zero>  
Onboard buttons not user accessible  
GPIO16 is NeoPixel onboard

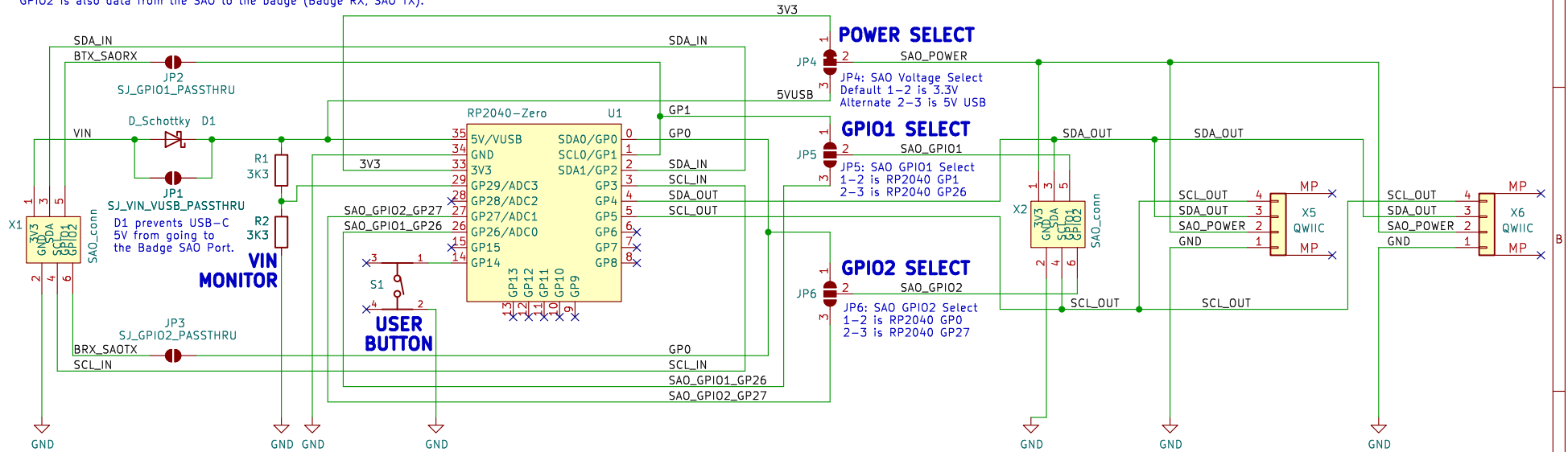
## TO SAO

SAO BADGE/HOST SOCKET  
Sullins SFH11-NBPC-D03-ST-BK female header  
<https://www.digikey.com/product-detail/en/sullins-connector-solutions/SFH11-NBPC-D03-ST-BK/59717-ND/4558818>

## QWIIC PORTS

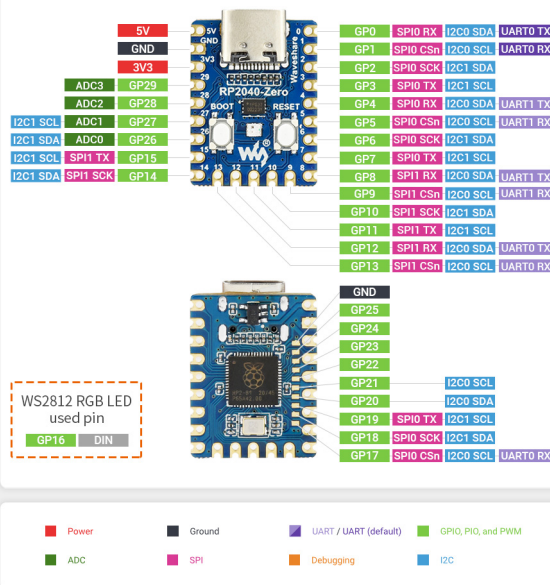
QWIIC SOCKET

Manufacturer: JST SM04B-SRSS-TB  
Digikey: 455-SM04B-SRSS-TBTR-ND  
LCSC: C160404



DEMO CONTROLLER FUNCTIONS  
INPUT 3-5V from Badge or USB-C  
GROUND  
3V3 regulated output to SAO  
VIN/2 input voltage measurement

SAO GPIO2  
SAO GPIO1  
User Button



DEMO CONTROLLER FUNCTIONS

SAO GPIO2 pass-through or Badge RX / SAO TX  
SAO GPIO1 pass-through or Badge TX / SAO RX  
SDA IN from Badge  
SCL IN from Badge  
SDA OUT to SAO  
SCL OUT to SAO

## COMMON I2C ADDRESSES

|                |         |      |
|----------------|---------|------|
| OLED (SSD1306) | 128x32  | 0x3C |
| OLED (SSD1306) | 128x64  | 0x3D |
| OLED (SSD1327) | 128x128 | 0x3C |

## SAO STANDARDS

PRIMARY: <https://docs.google.com/document/d/1EJqvkkLMAPsQ9VWF5A4e1WoiQMKyR5Giw5rqRmtmM/edit?usp=sharing>  
SUPPLEMENT: <https://hackaday.io/project/175182-simple-add-ons-sao>

All non-polarized capacitors are X7R or X5R ceramic unless otherwise noted.

Design by Andy Geppert © [www.MachineIdeas.com](http://www.MachineIdeas.com)

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